Regression modelling - anxiety predictors

Week-2

Task-1 (DS-1)

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1. Dataset Description:

The "depression_anxiety_data.csv" is used for regression modelling. This data is collected from university students having depression and anxiety.

id	school_year	age	gender	bmi	who_bmi	phq_score	depression_severity	depressiveness	suicidal	depression_diagnosis	depression_treatment	gad_score
1	1	19	male	33.333333	Class I Obesity	9	Mild	False	False	False	False	11
2	1	18	male	19.841270	Normal	8	Mild	False	False	False	False	5
3	1	19	male	25.102391	Overweight	8	Mild	False	False	False	False	6
4	1	18	female	23.738662	Normal	19	Moderately severe	True	True	False	False	15
5	1	18	male	25.617284	Overweight	6	Mild	False	False	False	False	14

anxiety_severity	anxiousness	anxiety_diagnosis	anxiety_treatment	epworth_score	sleepiness
Moderate	True	False	False	7.0	False
Mild	False	False	False	14.0	True
Mild	False	False	False	6.0	False
Severe	True	False	False	11.0	True
Moderate	True	False	False	3.0	False

Shape of dataset: (783,19)

```
(783, 19)
```

2. Data Cleaning:

The dataset contains missing values. So, missing values are removed from the dataset.

```
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   id                    783 non-null    int64  
 1   school_year           783 non-null    int64  
 2   age                   783 non-null    int64  
 3   gender                783 non-null    object 
 4   bmi                   783 non-null    float64
 5   who_bmi               783 non-null    object 
 6   phq_score             783 non-null    int64  
 7   depression_severity   779 non-null    object 
 8   depressiveness        780 non-null    object 
 9   suicidal              782 non-null    object 
 10  depression_diagnosis  782 non-null    object 
 11  depression_treatment  779 non-null    object 
 12  gad_score             783 non-null    int64  
 13  anxiety_severity      783 non-null    object 
 14  anxiousness           777 non-null    object 
 15  anxiety_diagnosis     779 non-null    object 
 16  anxiety_treatment     781 non-null    object 
 17  epworth_score         775 non-null    float64
 18  sleepiness            775 non-null    object 
```

After removing null values, the shape of data becomes:

```
(765, 19)
```

3. Data Preprocessing and Feature Engineering:

Some features are removed so that our Regression Model doesn’t learn noise. So, some features removed.

Removed Features:

- anxiety_severity
- anxiousness
- anxiety_diagnosis
- anxiety_treatment
- id

Reasons:

- These variables directly represented anxiety.
- They created target leakage with gad_score.
- Including them artificially increased model performance.
- Removing them improved model validity.

Removed Redundant Variables:

- who_bmi
- sleepiness
- depression_severity

But bmi, epworth_score, phq_score retained.

Reasons:

- Continuous numerical variables contain richer information.
- Categorized versions created redundancy and multicollinearity.

Numerical Feature Processing:

- StandardScaler() was used.
- Features were standardized before regression.

Categorical Feature Processing:

- OneHotEncoder(drop='first') was used.
- Dummy variable redundancy was avoided.

Train-Test Split:

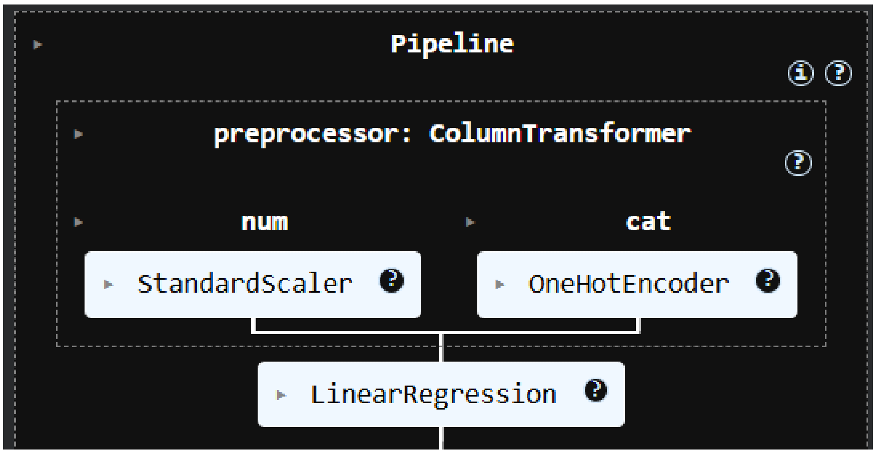
- Training data = 80%
- Testing data = 20%
- random_state = 42

4. Model Training:

Linear Regression model is used for regression modelling.

Pipeline components:

- Preprocessing pipeline.
- Feature scaling.
- One-hot encoding.
- Linear regression model.

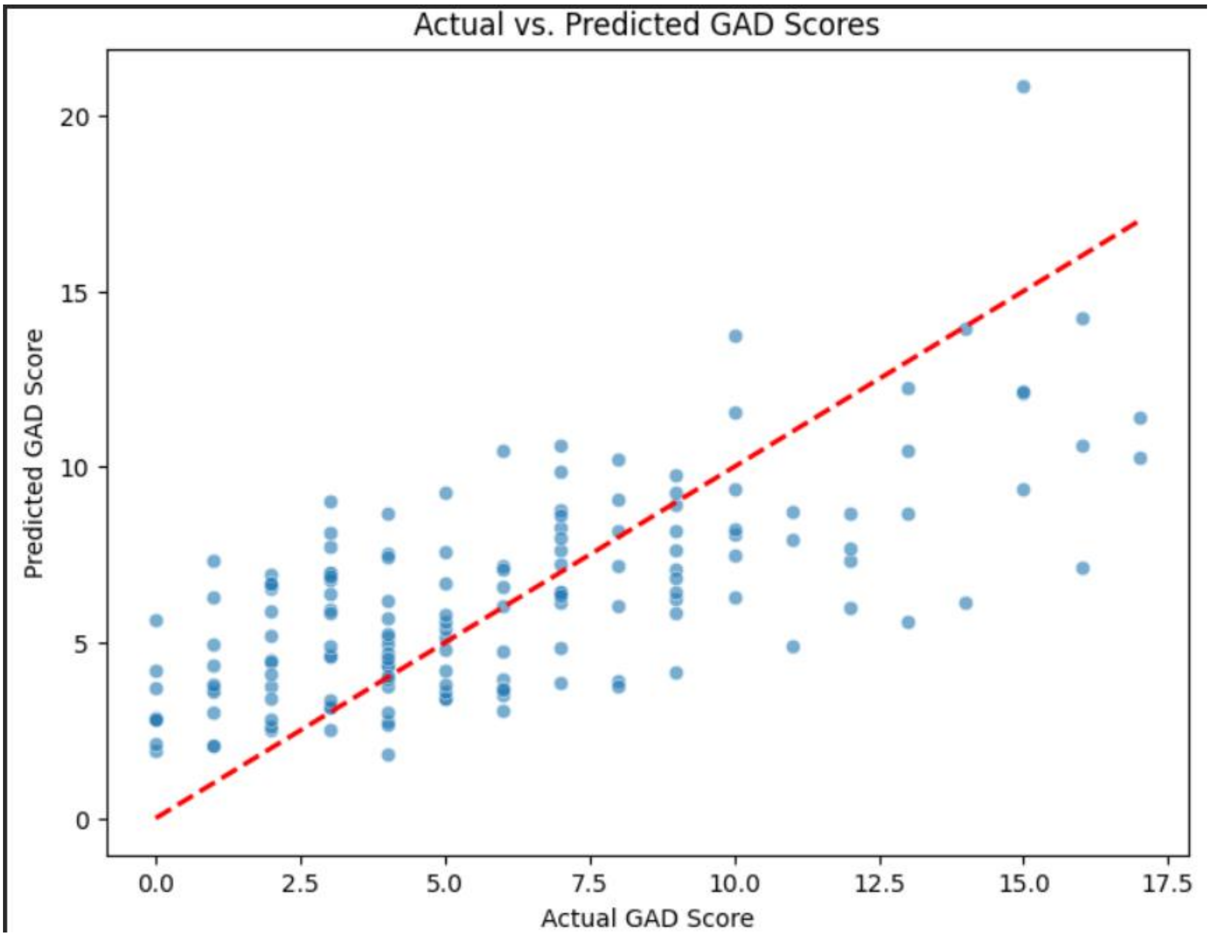


5. Model Evaluation:

Metric	Value
Mean Squared Error (MSE)	9.45
R Squared	0.46
Adjusted R Squared	0.42

```
Mean Squared Error: 9.45
R-squared: 0.46
Adjusted R-squared: 0.42
```

Visualization:



6. Regression Coefficients:

Feature	Coefficient
phq_score	2.929171
suicidal_True	0.731093
epworth_score	0.671137
depression_diagnosis_True	0.483831
age	0.286413
bmi	0.153060
school_year	-0.378550
depressiveness_True	-0.450033
depression_treatment_True	-0.601396
gender_male	-1.273413

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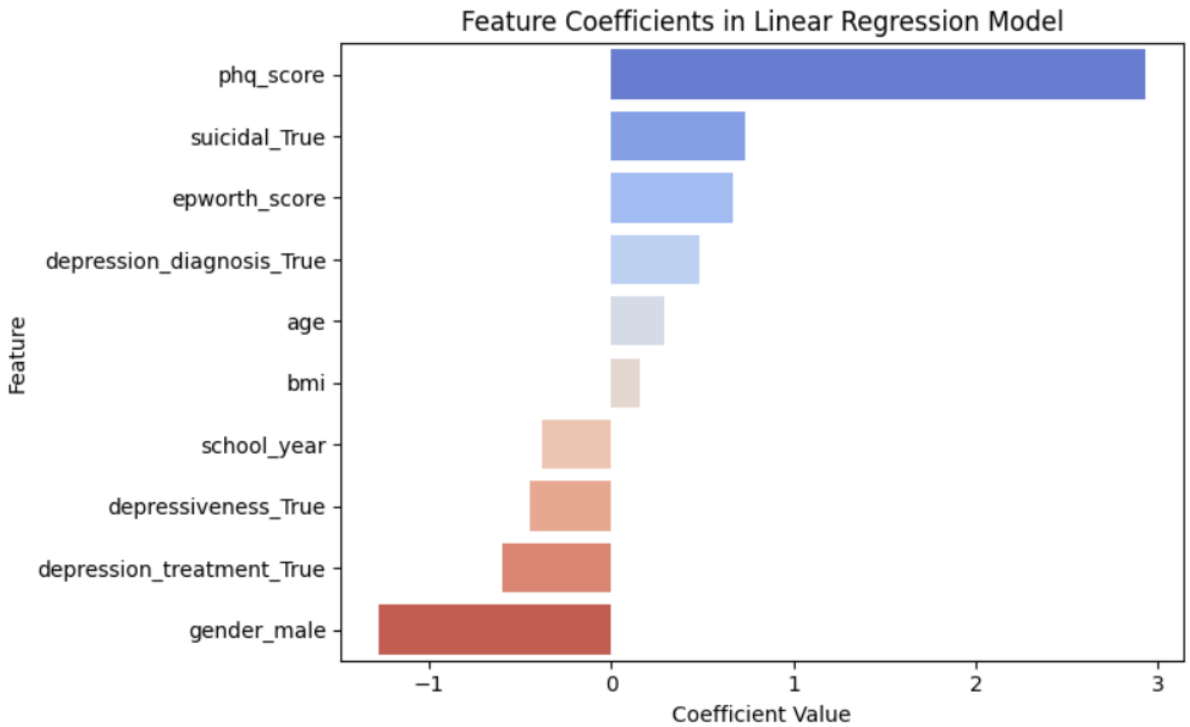

Positive Predictors:

- phq_score:
  - Strongest positive predictor of anxiety.
  - Higher depression scores increased GAD scores.
- suicidal_True:
  - Suicidal individuals tended to show higher anxiety.
- epworth_score:
  - Greater sleepiness/fatigue increased anxiety levels.
- depression_diagnosis_True:
  - Diagnosed depression was associated with higher anxiety.

Negative Predictors:

- gender_male:
  - Male participants showed lower predicted anxiety scores compared to females.
- school_year:
  - Higher academic year showed slightly lower anxiety levels.
- depression_treatment_True:
  - Participants receiving treatment showed relatively lower anxiety.

7. Final Conclusion:

- A linear regression model was successfully developed for predicting GAD-7 anxiety scores.
- Depression score (phq_score) emerged as the strongest predictor of anxiety.
- Sleep-related issues and suicidal tendencies also contributed positively to anxiety levels.
- Male gender and higher academic year showed negative relationships with anxiety.
- The final model achieved moderate predictive performance with an R^2 score of 0.46.
- Removing overlapping anxiety-related variables improved statistical validity and reduced multicollinearity.
- The final regression model was interpretable, statistically cleaner, and suitable for anxiety predictor analysis.